

Paulina Vvedenskaya

(408)-466-5876 | paulina.vvedenskaya@duke.edu | <https://www.linkedin.com/in/paulina-v/> | US Citizen

EDUCATION

Duke University, Pratt School of Engineering

Durham, NC

B.S. in Computer Science, Electrical and Computer Engineering

May 2028

- **Relevant Coursework:** Advanced Software Design, Intro to Applied ML, Advanced Data Structures & Algorithms
- **Activities:** The Cube (Entrepreneurship), Catalyst Tech Org., Club Tennis, Society of Women Engineers, Duke Applied ML

WORK EXPERIENCE

Esquire Deposition Solutions, LLC

San Francisco, CA

Software Engineering Intern

Jan 2026 – Aug 2026

- Implemented a lawsuit prioritization algorithm scoring 50,000+ active cases/day sourced using LexisNexis API data
- Designed a Salesforce-native LWC dashboard with custom Apex controllers that rank cases and tasks; provides AI-generated cold outreach emails; will save 80,000+ hrs/yr to increase revenue by \$2.75M across a 50+ rep sales team

Duke Office of Information Technology

Durham, NC

Software Engineering Intern — [Source Code](#)

May 2025 – July 2025

- Developed a Black Hole Routing site to manage 2M+ malicious IP addresses; implemented API token auth for security
- Engineered a PostgreSQL DB to efficiently cross-check IP lists and trigger block/unblock requests for 3k IPs in under 1s

PROJECT EXPERIENCE

AI Medical Billing Auditor — Python, Gradio, scikit-learn, ChromaDB, OpenAI API, OCR, Hugging Face

- Built an AI tool to extract CPT codes from uploaded bills via OCR + LLM-based NLP extraction w/ prompt engineering
- Compares fees to a 10k-row Medicare price db using a custom RAG pipeline w/ selectable hash/semantic embeddings
- Trained a random forest classifier with feature engineering on Reddit users and synthetic bills to predict risk (LOW/MED/HIGH, 0.995 macro F1); flags overcharges/duplicates; generates simple explanations and dispute letters

OOGASalad Game Creation Platform – Java, JavaFX

- Designed a data-driven no-code game authoring platform to create physics-based track games through a visual editor
- Led mid-project architectural pivot, redesigning the data model to separate reusable game catalogs from per-level configurations, requiring renegotiation of contracts across authoring, data, and engine layers
- Built full JSON pipeline (loaders, validators, registries) bridging game data to a physics engine w/ SAT collision detection

BLS Visa Appointment Monitor – Python, Selenium, 2Captcha API, SMTP

- Built an automation tool to monitor the BLS Spain visa appointment portal, solving image-grid CAPTCHAs via the 2Captcha API and navigating honeypot-aware, multi-step login/booking flows; sent real-time SMS alerts on availability
- Successfully secured an appointment in 24 days across ~575 automated checks, saving ~\$250 in visa expediting fees

Cell Society Cellular Automata Simulation Framework – Java, JavaFX

- Built an extensible, reflection-based rule factory supporting Game of Life, Spreading Fire, and agent-based simulations
- Designed a localized exception hierarchy decoupling model-layer errors from UI display across a 3-layer architecture

Hotel and Flight Price Aggregator – Python/Flask, React

- Building a Flask web app integrating Amadeus/Hotels.com APIs to aggregate trip costs across a group of travelers, compare total prices across dates and destinations, and receive automated email alerts on large price drops

LEADERSHIP AND INVOLVEMENT

- **Duke University The Cube Entrepreneurship Organization** – VP New Member Education, VP Social
- **FRC Robotics Team 971** – Captain, Alumni Mentor of 2x FRC Robotics Championship Div. Finalist team (top 0.5% globally)

Additional Information

Technical Skills: C++, Java, Python, JS/HTML/CSS, SQL, Flask, Git, Docker, REST APIs, JSON, React, Apex, Scikit-learn, PyTorch